

Literature, Data and Technology Review

Retail Location Intelligence | AI-supported market planning in Çankaya

Purpose: help café entrepreneurs shortlist candidate areas faster by comparing access to their intended customer surroundings and competition. Access uses indirect spatial indicators; time savings remain an unmeasured pilot goal.

Group 6 · Technical baseline: 58ce550 · Narrative updated 6 October 2026

1. Literature review: marketing context

The project supports café entrepreneurs and retail analysts during geographic market planning: selecting areas for on-site investigation before committing to a location. The research question is whether open spatial and demographic data can produce an explainable candidate shortlist and a separate comparison with existing café locations. The intended downstream audience includes nearby residents and visitors; demographic or behavioral customer segments are not observed.

Proposed business KPI: median time to prepare a candidate-area shortlist, compared with a manual map-based workflow. A pilot should give users the same area and task, record completion time, and review shortlist quality. No time saving or commercial improvement has been measured.

2. Thematic synthesis and research gap

Theme / source	Method and contribution	Implication for this project
Location analysis: Xiang et al. (2023) [1]	Starbucks coordinates and urban maps are used to examine spatial distribution and location factors. This is a descriptive preprint, not a Çankaya outcome study.	Supports examining local spatial context; it cannot establish the project’s weights or profitability.
Spatial validation: Mahoney et al. (2023) [2]	Simulations compare five CV approaches. Spatial grouping, especially with exclusion buffers, improves performance estimates for structured data.	Hold out entire neighborhoods; recognize remaining dependence across borders.
Operational example: Esri / Wendy’s [3]	GIS integrates demographics, store reports and site screening; the vendor describes sales analysis and cannibalization assessment.	A map can assist market planning, but this project lacks the sales records used in that example.

Together, the sources support a spatial decision workflow and careful validation. They do not justify transferring foreign location patterns or interpreting café presence as success. The capstone contributes an open-data, local prototype combining transparent MCDA rankings with a secondary Random Forest similarity layer. This is a practical implementation gap, not a claim of scientific novelty.

3. Literature conclusion

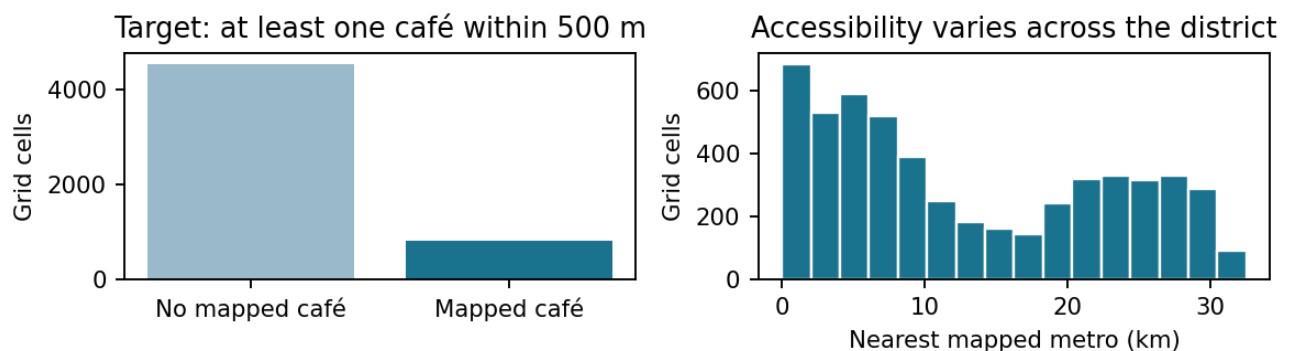
Use the ranking to narrow fieldwork, then check rents, pedestrian activity, customer fit and competition on site. Sales uplift, acquisition cost and revenue remain future outcome measures requiring additional data.

4. Data research: source and scope

Source	Verified scope / format	Role
OpenStreetMap [4, 5]	OSMnx / Overpass collection; district boundary, POIs and street networks. Raw GeoJSON; processed GeoParquet. Snapshot extraction date is not recorded.	Accessibility, amenities, café-presence target and competitive context.
Repository population CSV / TÜİK [6]	123 neighborhood rows; year=2025; total 939,486 across CSV rows. Original export provenance was not independently audited.	Residential population allocated by polygon-overlap area.
Committed feature grid [7]	5,361 clipped 300 m cells; 121 neighborhood labels; EPSG:32636; one cross-sectional snapshot.	Cell-level analysis and five-fold grouped validation.

5. Quality and exploratory findings

The committed grid has no duplicate cell IDs. Café presence is positive in 817 cells (15.24%); 4,544 cells are negative. This supports class-balanced RF training. These are cells with a café in a 500 m buffer, not counts of distinct businesses. Overlapping buffers mean nearby observations share information.



Mean café count per buffer is 0.69 (maximum 49); median café, restaurant and shop counts are all zero. Median nearest-metro distance is about 10.0 km. Commercial activity is uneven, so district-wide averages should not guide an individual site decision.

All 5,361 age-group values are missing; street names are missing in 3,417 cells (63.7%). The grid lacks office, government, kindergarten and nested café-competition features; stored RF scores are legacy outputs. Current code recomputes RF results. Updated school filtering requires explicit ISCED 2/3 tags, whereas the committed school counts are generic. OSM coverage and centroid-based buffers can misrepresent real walking access.

6. Privacy, suitability and conclusion

Inputs are public place data and aggregated population, not customer transactions or individual movement records. Aggregate population is a residential proxy, not observed footfall. The data is suitable for exploratory shortlist support; it does not establish customer demand, conversion or profit. Scores with unavailable penalties are labeled provisional, and missing demand layers are disclosed.

7. Technology review and comparison

Approach	Suitability / trade-off	Status
MCDA + GeoPandas	Inspectable contributions, adjustable scenarios and no commercial labels required. Weights are subjective; spatial proxies do not prove causal demand.	Implemented primary ranking.
Random Forest + grouped CV [8, 9]	Captures nonlinear tabular relationships; handles class weighting. Gini importance is descriptive and can favor correlated variables.	Implemented comparison layer.
Logistic regression	Simple baseline and coefficient interpretation; may miss nonlinear relationships without transformations.	Candidate baseline; not benchmarked.
Boosted trees	Potentially strong tabular predictor; extra tuning and validation effort.	Candidate; not benchmarked.
LLM recommendation	Could explain verified results in natural language, but adds cost and unsupported-answer risks without improving labels.	Not implemented or required.

8. Workflow, performance and practical example

OSMnx collects data, GeoPandas builds spatial features, scikit-learn computes similarity, and Streamlit/PyDeck presents maps and filters. MCDA ranks candidates for human review. RF separately estimates resemblance to mapped café areas. Corrected mean fold ROC-AUC is 0.960 on the committed grid [7]; this technical result does not demonstrate the proposed time-saving KPI.

The Wendy's example [3] demonstrates integration of maps and business reports for site screening. Our prototype adopts the screening workflow but cannot reproduce its sales-based analysis. Open tools avoid commercial API license fees; hosting, maintenance and field validation still require resources. Runtime training adds cold-start latency; the repository contains no measured latency or scalability benchmark.

9. Risks, opportunities and conclusion

Risks include incomplete OSM, outdated population, shared spatial buffers, uncalibrated similarity and misleading certainty from missing layers. Human review, coverage checks and provisional-score labels limit misuse. Priorities are a timed usability pilot, field checks of top candidates, buffer-separated spatial validation and verified rent/footfall data. Municipality data and popularity compatibility are planned, not active. The chosen stack fits an explainable student prototype; deployment at scale remains unverified.

References and implementation evidence

- [1] Xiang, Y., Chang, D., & Feng, X. (2023), Leveraging Urban Big Data for Informed Business Location Decisions (preprint). <https://arxiv.org/abs/2310.09778>
- [2] Mahoney, M. J., et al. (2023), Assessing the performance of spatial cross-validation approaches (preprint). <https://arxiv.org/abs/2303.07334>
- [3] Esri, Wendy's Deploys Esri Business Analyst for Site Selection and Predictive Modeling (vendor case). <https://www.esri.com/about/newsroom/announcements/wendys-deploys-esri-business-analyst-for-site-selection-and-predictive-modeling>
- [4] OpenStreetMap contributors, copyright and ODbL attribution. <https://www.openstreetmap.org/copyright>
- [5] Boeing, G. (2025), Modeling and Analyzing Urban Networks and Amenities with OSMnx; package documentation. <https://osmnx.readthedocs.io/en/stable/>
- [6] TÜİK, ADNKS population portal; repository CSV labels its observations as 2025. <https://nip.tuik.gov.tr/Home/Adnks>
- [7] Group 6, implementation baseline (README.md, data/processed, docs/model_validation.md); reviewed 6 October 2026. https://github.com/edasaruhan/SIC_AI_17_Capstone_Group_6/tree/58ce550c027d3ddce9f209fe3979bea849d5716a
- [8] scikit-learn, RandomForestClassifier documentation. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>
- [9] scikit-learn, StratifiedGroupKFold documentation. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html